

**LECTURE 19: MATH1061/7861**  
**TOPICS: PROPERTIES OF SETS, FUNCTIONS**  
**DEFINED ON GENERAL SETS**  
**DATE: TUESDAY APRIL 14, 2026**

Learning objectives:

- (1) Use the element method to prove properties of sets.
- (2) Use the laws of logical equivalence to prove two sets are equal.
- (3) Use set identities to prove two sets are equal.
- (4) Understand functions defined on general sets.

*\* MATH 7861 Assignment is due Friday  
April 24*

*\* Class Survey deadline is extended  
to the end of today*

Q1: Let  $A$  and  $B$  be subsets of a universal set  $U$ . Prove that

$$(A \cup B)^c \equiv A^c \cap B^c.$$

Recall  $A^c$  means:

$U - A$   
complement of  
 $A$  inside  $U$ .

$x \in (A \cup B)^c$  means

$$x \notin A \cup B$$

$$\Leftrightarrow$$

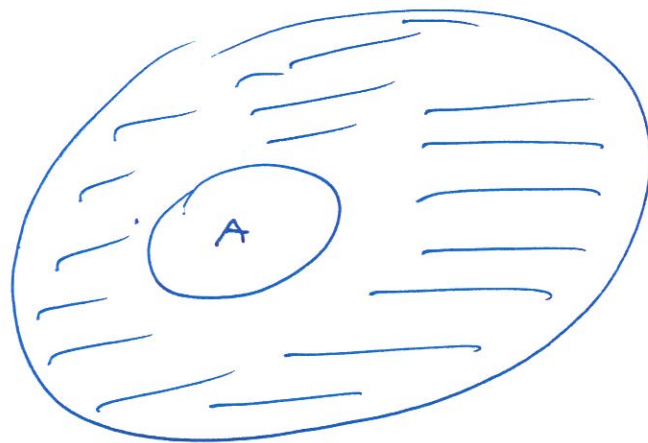
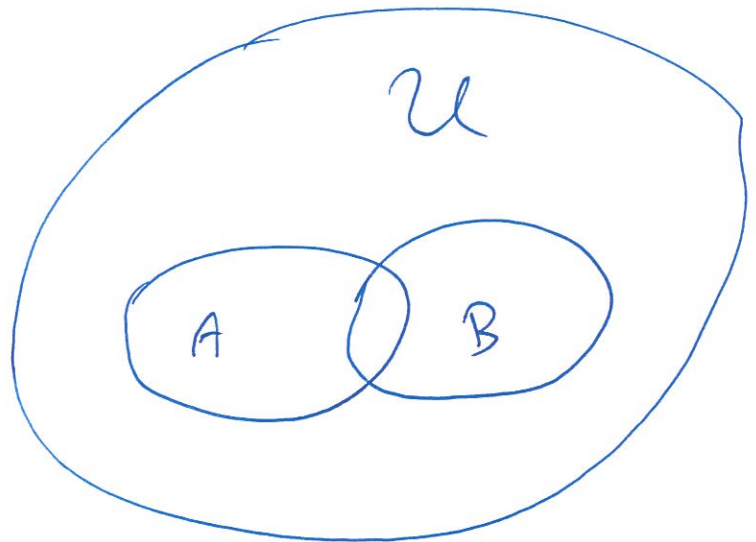
$$\sim (x \in A \cup B)$$

$$\Leftrightarrow$$

$$\sim (x \in A \vee x \in B)$$

$$\Leftrightarrow \text{De Morgan}$$

$$\begin{aligned} \sim (x \in A) \wedge \sim (x \in B) &\Leftrightarrow x \notin A \wedge x \notin B \\ &\Leftrightarrow x \in A^c \wedge x \in B^c \\ &\Leftrightarrow x \in A^c \cap B^c \end{aligned}$$



$A^c$  is  
the shaded  
area

**Set Identities** For all sets  $A$ ,  $B$  and  $C$  that are subsets of a universal set  $U$ :

- |                                       |                                                                                                              |
|---------------------------------------|--------------------------------------------------------------------------------------------------------------|
| 1. Commutative Laws                   | (a) $A \cup B = B \cup A$ (b) $A \cap B = B \cap A$                                                          |
| 2. Associative Laws                   | (a) $(A \cup B) \cup C = A \cup (B \cup C)$<br>(b) $(A \cap B) \cap C = A \cap (B \cap C)$                   |
| 3. Distributive Laws                  | (a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$<br>(b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ |
| 4. Identity Laws                      | (a) $A \cup \emptyset = A$ (b) $A \cap U = A$                                                                |
| 5. Complement Laws                    | (a) $A \cup A^c = U$ (b) $A \cap A^c = \emptyset$                                                            |
| 6. Double Complement Law              | $(A^c)^c = A$                                                                                                |
| 7. Idempotent Laws                    | (a) $A \cup A = A$ (b) $A \cap A = A$                                                                        |
| 8. Universal Bound Laws               | (a) $A \cup U = U$ (b) $A \cap \emptyset = \emptyset$                                                        |
| 9. <u>De Morgan's Laws</u>            | (a) <u><math>(A \cup B)^c = A^c \cap B^c</math></u> (b) $(A \cap B)^c = A^c \cup B^c$                        |
| 10. <u>Absorption Laws</u>            | (a) <u><math>A \cup (A \cap B) = A</math></u> (b) $A \cap (A \cup B) = A$                                    |
| 11. Complement of $U$ and $\emptyset$ | (a) $U^c = \emptyset$ (b) $\emptyset^c = U$                                                                  |
| 12. Set Difference Law                | $A - B = A \cap B^c$                                                                                         |

Exercise: Prove some of these using Laws of logical equivalence.

Q2: Use set identities to prove that

$$A - (A - B) = A \cap B$$

$$A - (A - B) = A - (A \cap B^c) \quad \text{Set difference}$$

$$= A \cap (A \cap B^c)^c \quad \text{Set difference}$$

$$= A \cap (A^c \cup (B^c)^c) \quad \text{De Morgan}$$

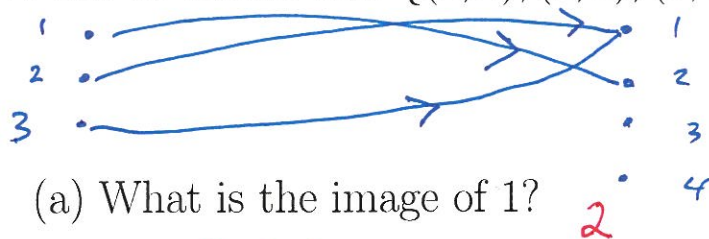
$$= A \cap (A^c \cup B) \quad \text{Double Complement}$$

$$= (A \cap A^c) \cup (A \cap B) \quad \text{Distributive.}$$

$$= \phi \cup (A \cap B) \quad \text{Complement Law}$$

$$= A \cap B \quad \text{Identity Law.}$$

Q3: Let  $X = \{1, 2, 3\}$  and  $Y = \{1, 2, 3, 4\}$ .  
 We saw in videos that  $\{(1, 2), (2, 1), (3, 1)\}$  is a function.



(a) What is the image of 1?

$$f(1) = 2$$

(b) What is the image of  $\{2, 3\}$ ?

$$\{1, 1\} = \{1\}$$

(c) What is the inverse image of 2?

$$f^{-1}(2) = \{1\}$$

(d) What is the pre-image of  $\{3, 4\}$ ?

$\emptyset$

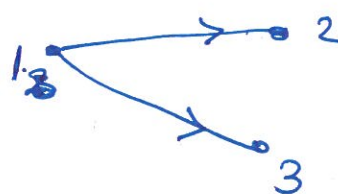
(e)

Is

$$\{(1, 2), (2, 2), \cancel{(1, 3)}\}$$

a function  $X \rightarrow Y$ ?

No.



3 is not mapped to anything.